



Society of Petroleum Engineers

SPE-227639-MS

Edge Generative AI Solutions for Industrial Security

A. Alangari, F. Alsumail, A. Dajani, Ahmad A Alkhabbaz, and A. M. Alanazi, Saudi Aramco

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227639-MS](https://doi.org/10.2118/227639-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

As computing chips become smaller and more efficient, the amount of compute that can be processed by smaller devices are getting increasing exponentially. This, coupled with the industrial need to enhance security, safety, and operational efficiency through Artificial Intelligence (AI) in remote areas where there are limited connectivity options have created unforeseen opportunities at the edge, minimizing the dependence on highspeed connectivity. This white paper presents a comprehensive analysis of the industrial security applications of Edge Generative AI (Gen AI), a novel technology that enables real-time processing of AI models at the edge of industrial networks where users, and data points are close to each other.

The industrial security sector is particularly well-suited to benefit from Edge Gen AI due to its reliance on complex, distributed systems and high-stakes decision-making that require real-time data. By deploying AI models closer to the source of the data, Edge Gen AI enables faster processing, reduced latency, and improved real-time insights – all of which are critical for enhancing security, detecting potential threats, and responding to incidents. Furthermore, the integration of Edge GenAI with Internet of Things devices and sensors can unlock new levels of monitoring, automation, efficiency in areas such as threat detection, incident response, and security analytics.

This white paper provides an in-depth examination of the technical architecture, methodology, and performance results of Edge Gen AI in industrial security applications, including Gen-AI enabled Security and Surveillance, Autonomous Drones, and Emergency Response. Our goal is to share the knowledge gained from this pioneering technology and provide valuable insights for other organizations seeking to harness the power of Edge Gen AI to enhance their industrial security posture.

Introduction

As computing chips become smaller and more efficient, the amount of compute that can be processed by smaller devices are getting increasing exponentially. This, coupled with the industrial need to enhance security, safety, and operational efficiency through Artificial Intelligence (AI) in remote areas where there are limited connectivity options have created unforeseen opportunities at the edge, minimizing the dependence on highspeed connectivity. This white paper presents a comprehensive analysis of the industrial security applications of Edge Generative AI (Gen AI), a novel technology that enables real-time processing of AI models at the edge of industrial networks where users, and data points are close to each other.

The industrial security sector is particularly well-suited to benefit from Edge Gen AI due to its reliance on complex, distributed systems and high-stakes decision-making that require real-time data. By deploying AI models closer to the source of the data, Edge Gen AI enables faster processing, reduced latency, and improved real-time insights – all of which are critical for enhancing security, detecting potential threats, and responding to incidents. Furthermore, the integration of Edge GenAI with Internet of Things devices and sensors can unlock new levels of monitoring, automation, efficiency in areas such as threat detection, incident response, and security analytics.

This white paper provides an in-depth examination of the technical architecture, methodology, and performance results of Edge Gen AI in industrial security applications, including Gen-AI enabled Security and Surveillance, Autonomous Drones, and Emergency Response. Our goal is to share the knowledge gained from this pioneering technology and provide valuable insights for other organizations seeking to harness the power of Edge Gen AI to enhance their industrial security posture.

Why Generative AI?

Conventional AI employs established statistical and computational algorithms to examine and derive insights from data. Conversely, **Generative AI** builds upon this foundation by synthesizing novel data that emulates the properties of real-world datasets, thereby expanding AI-driven analytical capabilities, Ravichandran et al [1]. Gen AI includes all use cases where new data generation is required by building on top of existing data which may include all sources of past data, including maintenance logs, sensors historical data to establish the context, and integrating different forms of current data such as current sensor data, cameras, and other IoT data points. We will go over some example use cases in details in this paper such as Gen-AI powered surveillance, and Gen-AI powered emergency response.

Why deploy on the edge?

Given the advancements in chip-making processes, the increasing computation density of AI components, led to them being increasingly considered to overcome difficult connectivity environments in remote sites. This includes high latency, low bandwidth, and critical data management considerations, which Edge AI deployments can resolve as follows:

- Minimize latency
 - Enable real-time monitoring, alerting, and warning, reducing the communication latency by running the AI model in the same site where the data is generated, rather than taking a round trip to the data center to process the data.
- Minimize bandwidth (data rate)
 - Minimize the bandwidth utilization of AI models by only transmitting necessary data, with the majority of the data and compute happening on the site itself. This includes physical bandwidth limitations such as the one we are anticipating in the industrial 450 MHz LTE network in Saudi Arabia, as well as cost savings related to reducing bandwidth consumption on leased terrestrial and non-terrestrial links
- Cybersecurity Enhancement
 - Enhance the cybersecurity of deployed solution by mitigating the amount of critical site data being transmitted through the network, as the data processing and storage happens mostly on the edge.
- Improved Resiliency

This is critical when the core network is disconnected for any unforeseen reason, the devices themselves, or the Local Area Network (LAN) will be able to respond to these interruptions, manage AI workloads locally, and continue to transmit warnings to the local users in the edge regardless of data center availability.

This paper includes a deep dive on Edge AI, its potential use cases, and how it integrates with Gen-AI to generate new, aggregated site intelligence from the data residing in segregated monitoring tools, with a special focus on conducted tests and metrics related to Industrial Security.

Methodology and Approach

In the development and deployment of AI-driven solutions within industrial security environments, a rigorous and robust methodology is essential to ensure technical feasibility, operational efficiency, and scalability. This section outlines the methodologies employed to architect, develop, and deploy Edge AI technologies across Industrial Security verticals. The methodologies were carefully designed to progressively enhance the solution's capabilities, starting from foundational deployments and scaling to more complex integrations and continuous refinement. The specific methodologies used in the testing

1. Initial Proof of Concept (PoC): Minimum Valuable Product (MVP)

- The deployment strategy began with the implementation of a Minimum Valuable Product (MVP) to establish foundational capabilities. The MVP focused on integrating existing non-AI-powered devices, such as cameras and sensors, to demonstrate the feasibility of Generative AI-Powered Security Surveillance. This initial deployment was conducted in a completely isolated Quality Assurance (QA) network to ensure controlled testing and performance evaluation. The objectives of the MVP were multifaceted:
 - To validate the technical feasibility of connecting legacy devices to AI-driven systems.
 - To assess the performance of the integrated solution under real-world conditions.
 - To identify areas for improvement in terms of functionality, efficiency, and scalability.
- The successful execution of the MVP not only validated the technical viability of the solution but also fostered trust and collaboration between Aramco and the vendors. This trust became a cornerstone for expanding the solution to address the broader use cases outlined in this paper.

2. Collaborative Development and Enhanced PoC

- Building on the success of the MVP, a joint taskforce was established between domain experts from Aramco and technology experts from the vendors. This collaborative effort was instrumental in refining the MVP and advancing the deployment of the second PoC. The taskforce focused on:
 - Incorporating feedback and lessons learned from the initial deployment to enhance the solution.
 - Adapting the solution to meet enhanced performance metrics and acceptance criteria.
 - Expanding the scope of the PoC to include additional use cases and more complex integration requirements.
- This phase underscored the importance of iterative development and the synergy gained from combining domain knowledge with technical expertise, ensuring the solution aligns with both business and technical objectives.

3. Continuous Monitoring and Improvement

- To ensure the solution remained aligned with evolving requirements and performance expectations, a continuous monitoring and feedback framework was implemented. This involved:

- Regular assessment of system performance, including metrics related to accuracy, response time, and reliability.
- Regularly collecting feedback from end-users and stakeholders to identify areas for improvement.
- Conducting iterative refinement cycles to address identified gaps and optimize the solution.
- Simultaneously, the deployment process was scaled incrementally, ensuring a smooth transition to production readiness while maintaining rigorous quality control standards. This approach allowed for continuous improvement and adaptation to dynamic requirements, leveraging agile development cycles to keep pace with changing demands.

4. Structured Methodology for Edge AI Deployment

- The overall methodology adopted for deploying Edge AI technologies across Industrial Security business case verticals included:
 - Leveraging cutting-edge Generative AI models (e.g., Llama 3) and Computer Vision models (e.g., YOLOv8) to ensure superior performance and accuracy.
 - Developing and deploying integrated software solutions to enable a user-friendly experience, and enable the management of edge devices and models.
 - Conducting rigorous testing and validation processes at each stage of development to ensure robustness and reliability.
- This structured approach ensured that the deployment of Edge AI technologies was both methodical and efficient, minimizing risks and maximizing the likelihood of successful implementation.

5. Real-World Testing and Threat Detection

- The integrated Edge AI solution was subjected to real-world testing in industrial security environments. The system was tasked with detecting, analyzing, and responding to actual threats in real time. Performance monitoring was conducted continuously to evaluate the effectiveness of the AI-driven security framework, focusing on threat detection accuracy, response time, and system reliability. This phase demonstrated the solution's ability to handle real-world challenges, showcasing its practical value and operational impact.

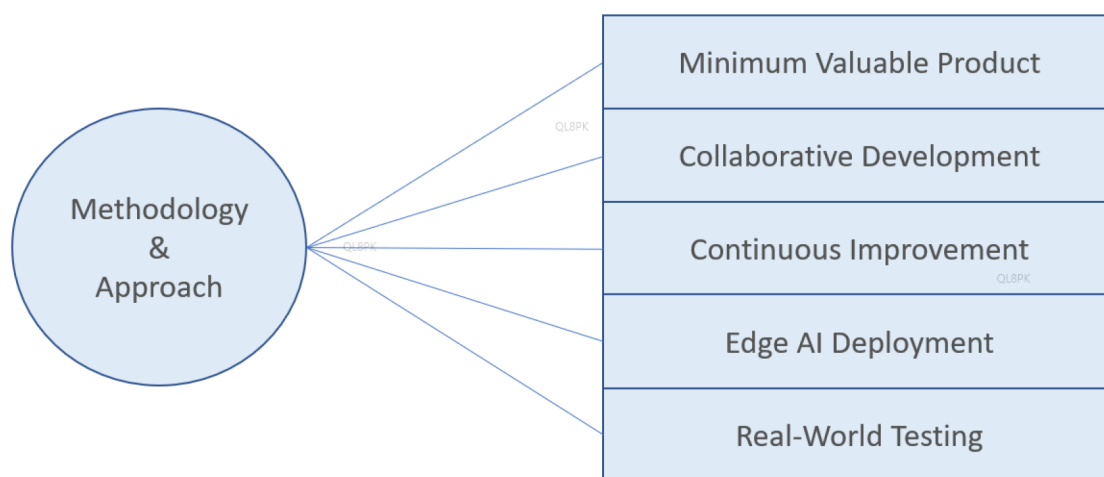


Figure 1—Methodology and Approach.

Design and System Components

The PoC focused on three primary use cases:

1. Gen-AI enabled Security and Surveillance

AI-powered cameras and sensors were deployed to track people and objects, detect anomalies, and trigger alerts in real-time all on site, with the possibility to convert normal cameras into smart cameras with an AI Box.

The solution integrates several advanced components:

A. Cameras:

These are the regular IP that can be equipped with AI processing via the Edge Box, this allows them to utilize Computer Vision (CV) algorithms enabling real-time detection and recognition of people, vehicles, events, and more.

B. Edge AI Boxes:

For cameras that do not have processing power in them the Edge AI Box will take the streams from these cameras and process them on site locally, this setup allows CV models and algorithms to be able to run on several of these streams. Furthermore, all the detections, alerts, and streams are saved locally on the Edge AI Box.

C. Video Management System (VMS):

The Video Management System (VMS) is responsible for managing the video feeds from all cameras. It retrieves, archives, and deletes content based on predefined configuration and retention rules. Installed on on-prem compute, the VMS ensures that all video data is handled efficiently and securely.

D. Edge AI Servers:

This is an Edge Computing device that is utilizing AI Accelerator cards to run AI models. This device runs a Large Language Model (LLM) and is available on site in order to provide completely private, secure, and air-gapped solution, thus eliminating the need for cloud connectivity to run the LLM. Furthermore, this LLM is linked with the meta-data that we get from the Edge AI Box and Smart Cameras in order to search and retrieve information about the camera feeds.

E. Security Center:

The Command Center is the hub where all the metadata is stored and is the interface between the users and the results of the model, it provides the following features:

- LLM Chat Interface: It utilizes the available edge compute to run user queries in order to get information in a natural language, resulting in a reduction of time and effort to retrieve information.
- Events and Alerts: The system will be able to automate the detections of specified events by the user, it would provide the timeframe and the video playback of the event.
- Analytics: The system can be used to analyze the behavior of the people and collect data for optimization purposes.

2. AI enabled Autonomous Drones

AI-enabled Autonomous Drones are advanced drones that is powered by an operational management platform. The platform is connected to drones, cameras, and sensors to allow the automation of detecting incident and deployment of the drones to the incident location. Furthermore, the drones are equipped with state-of-the-art systems that allows it to avoid obstacles, manage its battery, communicate with other drones, and return to charge without manual intervention. Moreover, the drones can be configured to include no-fly zones and areas to avoid. Additionally, the platform will

be able to monitor the entire drone systems in collaboration in order to achieve the goal of collecting information about the incident.

The solution contains the following:

A. Drone Command Center:

The Drone Command Center (DCC) is the main software platform which the users interact with and oversees and controls the entire drone fleet. It dispatches drones on mission, manage communication between drones for coverage, and the collection, processing, and storage of video streams. Furthermore, the system supports mission planning, incident response, and generates reports that include all data collected from the flight. This DCC is usually deployed on a lightweight computer within the site of the drones.

The User Interface of the DCC shows all active and idle drones on a live map, with the option override the autonomous flight and remotely control it when needed. In addition, it provides integrated AI capabilities, both on-drone and on local server which enables CV algorithms and model run on the streams for analysis purposes, which helps in assessing the threats and take informative-decision making. Lastly, the system can process fixes security cameras to detect incidents and asses of there is a need for drone deployment.

B. On-prem Compute:

This on-prem compute will include the full software stack that is required for the AI-powered use cases, including all the computer vision and LLM models that is required to operate the system. This edge compute will convert alerts, triggers, and LLM-based commands to commands that the DCC will be able to understand and command the drones accordingly.

C. Video Management System

This device is used to store the videos for long-term, and is deployed in order to comply with internal and national data??? regulations. The videos are stored in the VMS for a minimum of 90 days, and the size of which can be expanded, as required by operations by deploying additional storage devices.

D. AI-powered Drone

This drone will include autonomous navigation capabilities given the commands it receives from the Drone Command Center. This will include the dispatchment to the location, avoiding any obstacles on the way, streaming the video through available connectivity, including Wi-Fi, and Private/Public cellular networks, and are equipped with industrial transmitters to enable multiple SIM card options are available in parallel, including different Internet Service Provider SIM cards, to ensure the reliable communication. This connectivity is prioritized based on the lowest latency to the edge Compute and the edge-deployed drone command center.

E. Cameras

Cameras are deployed and integrated with the edge compute in order to trigger AI-based emergency response applications based on security, safety, and operational pre-defined triggers and warnings and trigger the DCC to react accordingly.

3. Gen-AI enabled Emergency Response

This use case will combine the 2 use cases mentioned above where the whole integrated system comes to life, triggering the Emergency Response Teams, including the AI-enabled autonomous drones based on the inputs from the AI-enable Security Surveillance Systems, as well as the integration with the existing site sensors. This use case includes Gen-AI capabilities to trigger the proper response with the necessary documentation and logging of the critical events, maintaining the logs of the communication through voice-to-text conversion, as well as generating the report of the incident. The below will go into details of the novel system components related to this solution.

A. Site Sensors

Site sensors are regular IoT devices with little to no AI capabilities, which will be connected to the AI box to add an additional layer of monitoring and confirmation prior to triggering the emergency response through multi-modal AI models taking into consideration of the video feeds and the sensor input.

B. Edge AI Boxes

In addition to enable non-AI powered cameras, it will include the sensor readings from the site sensors.

C. Edge AI Servers

This device runs the Large Language Model (LLM) and is available on site in order to provide completely private, secure, and air-gapped solution, thus eliminating the need for cloud connectivity to run the LLM. Furthermore, this LLM is linked with the meta-data that we get from the Edge AI Box, Smart Cameras in order to search and retrieve information about the camera feeds and sensor data. This will also be used to run speech detection models for voice-to-text conversion and generating incident reports.

D. Security Center

The security centers act as the main dispatching entity where the Gen AI proposes the correct entities to alert and the actual alerting to the emergency response teams will occur. The emergency response drones will request the security center's approval prior to the flight, and will trigger the flight based on active integration between the on-prem compute and the Drone Command Center.

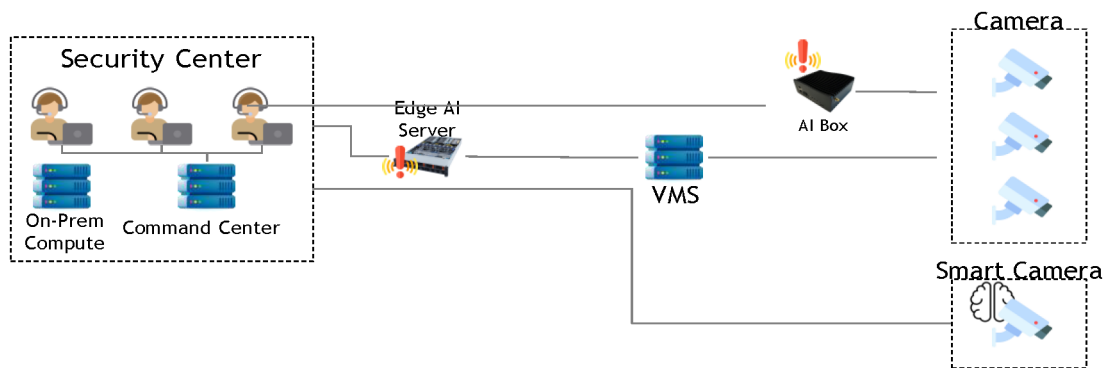


Figure 2—Solution Setup of Gen-AI enabled Security and Surveillance.

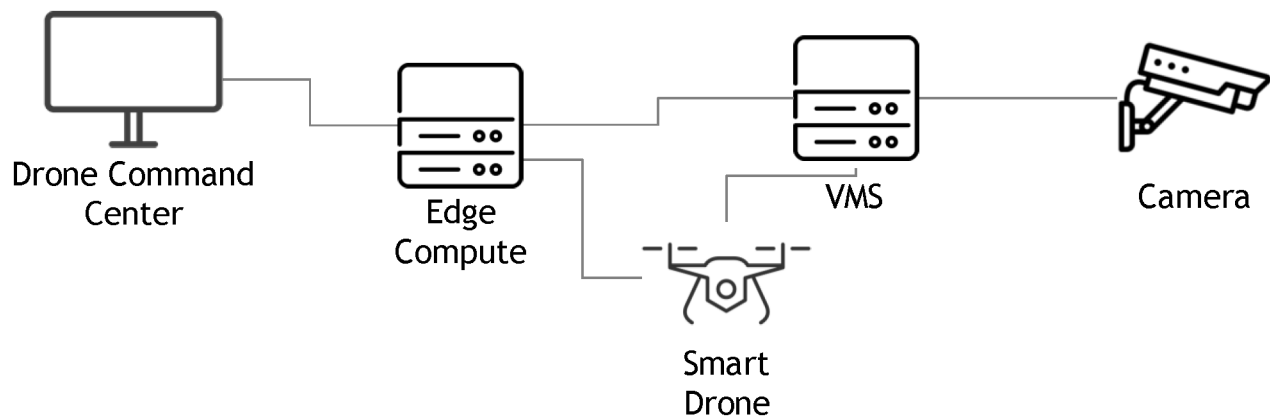


Figure 3—Solution Setup of AI-enabled Autonomous Drones. TODO remove background

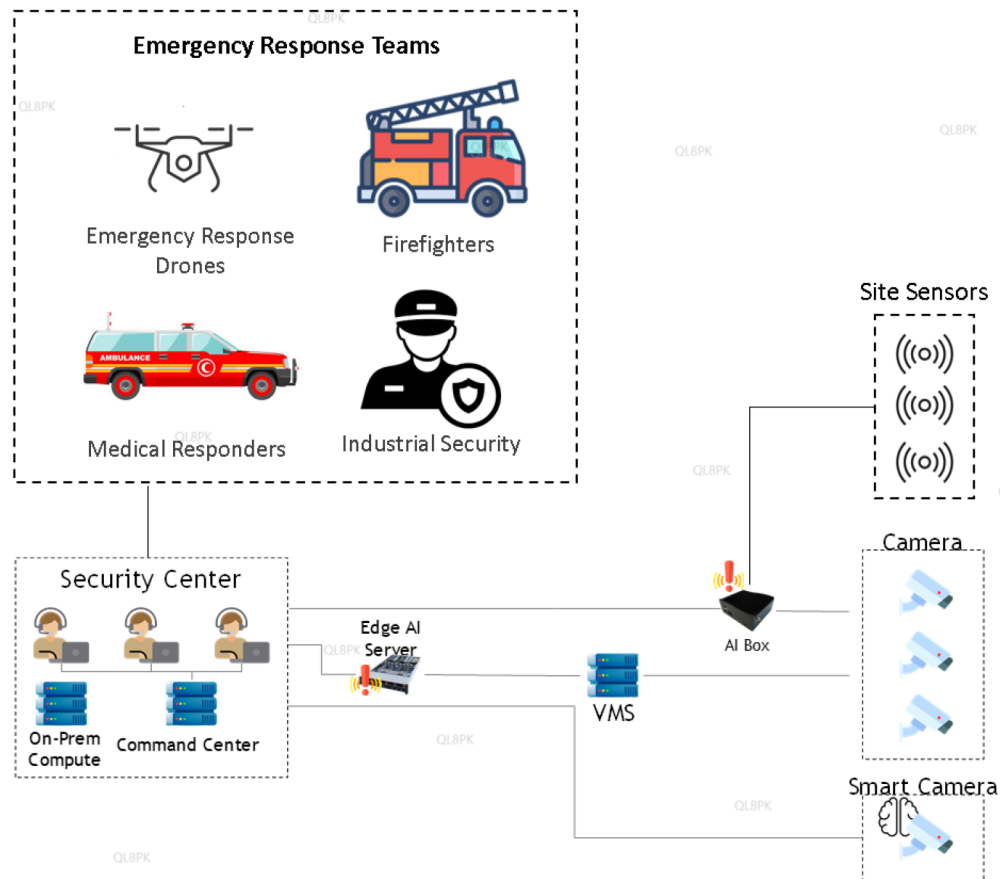


Figure 4—Solution Setup of AI-enabled emergency response

Use Cases

1. *Gen-AI enabled Security and Surveillance:*

Gen-AI enabled Security and Surveillance is capable of delivering several features, including Intrusion Detection, Access Control Monitoring, Track & Locate across cameras, People Analytics, Weapon Alert, Heatmaps, Gen-AI queries. It will also increase the accuracy of the system, which will result in less false positives caused by existing systems.

In addition, it could perform the following scenarios:

- Cross camera tracking (people and activities)
- Locating individuals
- Gen AI based Active Queries
 - When did person X passed by area Y?
 - Where did person X go after this time (HH:MM:SS MM-DD-YYYY)?
 - How many bags were left behind for longer than 15 minutes
 - What time did Person X come to this facility?
 - Where is Person X?
- Gen AI based daily reports including detailed and summarized safety and security reports.
- SLM-powered, AI-based triggers

- Person waving hands – signaling distress
- Person accessing restricted areas
- Person fainting
- Person waving hands (Emergency signal)
- Person not wearing proper PPE in restricted area

2. *AI-enabled Autonomous Drones:*

AI-enabled Drones fly autonomously to an incident location and stream livefeed video to the command center to achieve better situational awareness and faster incident response. This use case can perform the following scenario:

- Fire detection based on IoT readings will trigger an Autonomous Drone Flight Mission on the Command Center.
- After mission approval, the drone will fly autonomously to the location of the incident.
- Drone flight path on a map and video from camera on the drone streams to the Drone Command Center to enable situational awareness for the user.
- Drone Handover is enabled to support longer mission times in case Drones need to recharge.
- Drone control using natural language for faster response time rather than using a controller.
- User request drone through LLM to "Move 50 meters north" command to get better visibility.
- [Figure 6](#) resembles further use cases to be performed utilizing edge AI on fixed-wing drones.

3. *AI-powered Emergency Response*

AI-powered emergency response enables triggering the emergency response after detecting an incident on site cameras and sensors. This emergency response system will ensure that all the critical logs are taken into place to be reviewed after the incident to ensure compliance, as well as ensuring that the proper emergency response procedures are taken into place with maximum efficiency to minimize the damages of the incidents. This includes the below possible scenarios

- Site IoT (Cameras & sensors) detects security incidents (e.g., intrusion) and will trigger the security center to dispatch the emergency response drones and the industrial security personnel
- Site IoT detects fire in one of the facilities, and trigger the emergency response through the security center to dispatch firefighting personnel
- Cameras detect a fainting incident and triggers the emergency response for the medical responders through the security center
- Generate a log based on the communication between the security sensors and the emergency responders
- Generate a report on the incident, giving the start date & time, end date & time, communication log, proposed procedure, implemented procedure, equipment damages, injuries & fatalities

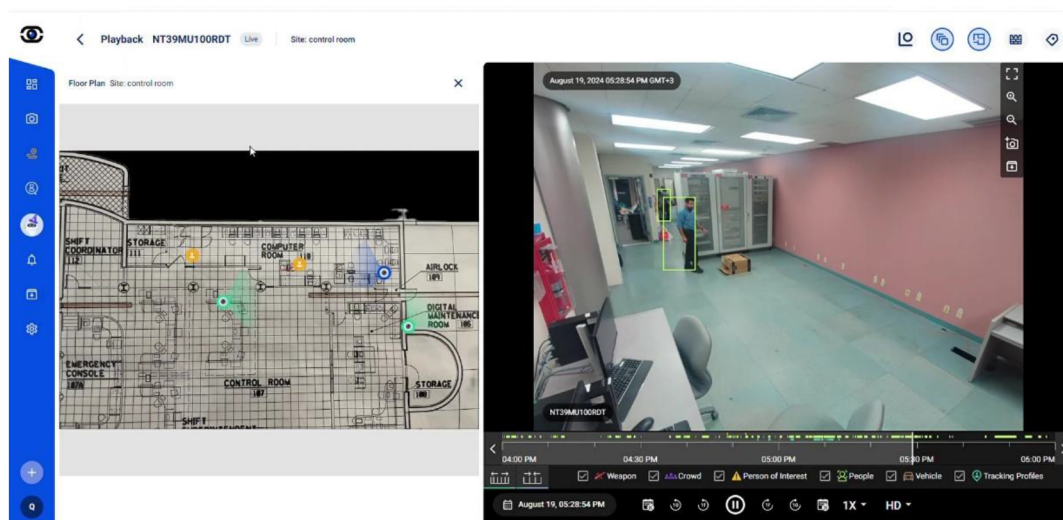


Figure 5—Results of Gen-AI enabled Security and Surveillance

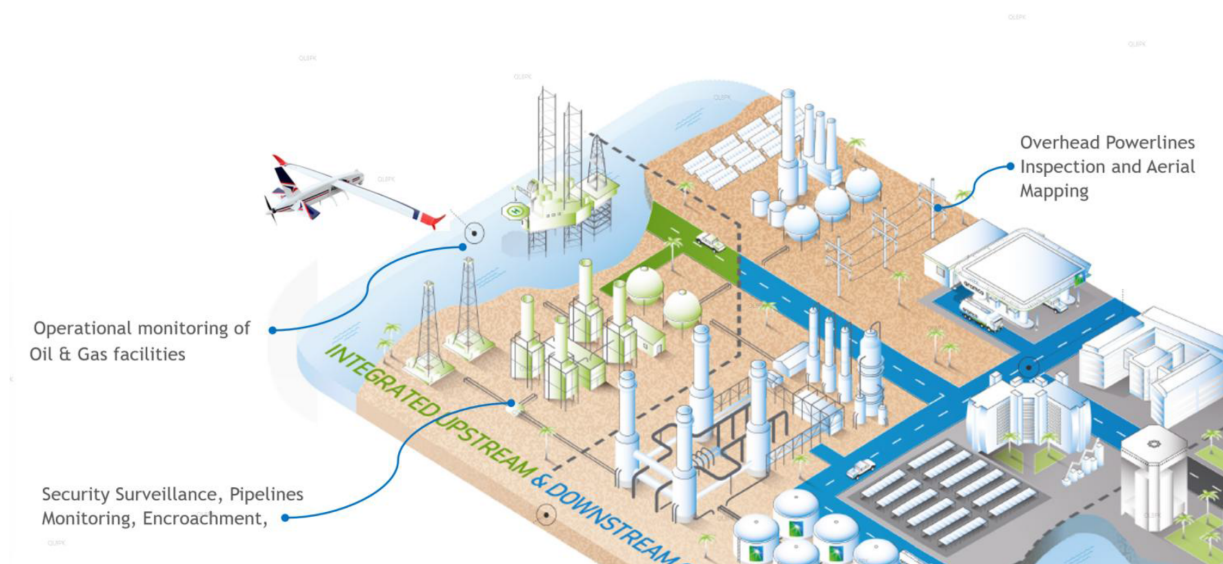


Figure 6—Example of AI-enabled Autonomous Drones

Performance Results

The integration of Gen-AI enabled Security and Surveillance systems, AI enabled Autonomous Drones, Gen-AI powered Emergency Response Systems has yielded promising results, demonstrating the potential of Industrial Security use cases in driving operational efficiencies, enhancing safety, and boosting productivity in industrial settings.

Operational Efficiency

The use of Gen-AI enabled Security and Surveillance systems has been shown to improve operational efficiency in various industries. By leveraging edge Gen AI, we can optimize our security protocols, minimizing unnecessary responses to false alarms, and patrolling activities, and improving response times to actual incidents, ultimately leading to significant reduction in manual monitoring times.

Improved Emergency Response Time

The integration of AI-powered cameras and sensors in industrial settings has been shown to have a positive impact on emergency response times by detecting anomalies and triggering alerts in real-time. Additionally, AI-powered emergency response systems can lead to a 35% faster response time, according to the US Cybersecurity and Infrastructure Security Authority [2], in our case, this reduction was mainly due to the instantaneous detection and reporting of incidents as opposed to traditional methods. Furthermore, Gen AI enhanced the emergency response time by 30% after only 3 months of adoption, and we are seeing this start to develop as end users get used to the AI systems deployed in place and test its capabilities. Ultimately, both conventional AI and Generative AI have positive effects on the response time, where both could be deployed and work hand-in-hand to improve current security, safety, and operational systems.

Recommendations

Based on the successful outcomes and observed challenges, the following recommendations are proposed:

- Expand the deployment of edge Gen-AI enabled Security and Surveillance systems, AI-enabled Autonomous UAVs, and emergency response systems in different sites in Aramco
- Explore and test further industrial security use cases, and the utilization of various types of AI Models to enhance operational efficiency and safety in different industrial security verticals
- Invest in continuous collaboration and innovation with leading technology partners to stay at the forefront of digital transformation.
- Explore utilizing the Edge AI infrastructure in a shared services manner where applicable, enabling both security, and core operations growth.

Key Takeaways and Conclusion

The integration of Gen-AI enabled Security and Surveillance systems, AI-enabled Autonomous Drones, Gen-AI powered Emergency Response has the potential to transform industrial security operations. By leveraging AI and machine learning technologies, companies can achieve significant gains in operational efficiency, safety, and workforce productivity. Key highlights from these use cases include:

- Real-time monitoring and analysis of security-related data reduces the risk of security breaches and improves response times.
- Optimized security protocols and reduced unnecessary responses to false alarms increase operational efficiency and productivity.
- AI-powered systems improve industrial safety and minimize accident risks.
- Gen-AI empowers security personnel to focus on complex tasks, boosting overall productivity and job satisfaction.

References

1. Ravichandran, Prabu & Rachakatla, Sareen & Machireddy, Jeshwanth Reddy. (2023). Data Analytics Automation with AI: A Comparative Study of Traditional and Generative AI Approaches.
2. Bono, J., Grana, J., and Xu, A., "Generative AI and Security Operations Center Productivity: Evidence from Live Operations", arXiv e-prints, Art. no. arXiv:2411.03116, 2024. doi:[10.48550/arXiv.2411.03116](https://doi.org/10.48550/arXiv.2411.03116).